

When Predictive Surrogates Fail as RL Environments:

A Calibrated Physical Twin as Soft Regularizer for HVAC Control

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Abstract

Reinforcement learning controllers for HVAC systems are typically trained on neural-network surrogates because high-fidelity simulators are too slow for the millions of environment steps consumed by modern policy-gradient methods. A natural assumption is that a more physically faithful surrogate produces a better training environment. We test that assumption on the BOPTTEST bestest_air testcase and report a negative result: a calibrated physical twin with explicit zone thermal capacitance reaches a 24-hour rollout RMSE of 0.64 C versus 1.47 C uncalibrated, yet fails as a stand-alone RL training environment with live closed-loop RMSE above 4 C. We resolve the gap by using the calibrated twin as a frozen soft physical regularizer for a smoother control-oriented surrogate. The canonical hybrid backend sustains 1,786.8 environment steps/s on one CPU thread, an 85.0x speed-up over the live BOPTTEST RTE HTTP loop under the same 15-min protocol, and restores live closed-loop RMSE below 0.65 C. The optimal regularization strength is controller-family specific: $\lambda_{\text{temp}}=0.10$ for thermostatic PPO, but $\lambda_{\text{temp}}=0.00$ for HDRL and 17D MORL. A pre-registered transferability block further shows that the Stage A/B/C inverse surrogate-calibration pipeline transfers across three hydronic BOPTTEST testcases with 60.2-87.8% RMSE_T improvement and consistent C_{zon} re-identification near 1.9x bestest_air, whereas frozen controller transfer is regime-dependent and not deployment-ready without target-specific fine-tuning. All numerical justifications are provided in Supplementary Tables S1-S11.

Keywords: HVAC control; deep reinforcement learning; digital twin; physics-informed machine learning; BOPTTEST; multi-objective reinforcement learning.

1. Introduction

The paper is organized around a falsifiable question: does improving predictive fidelity of a building surrogate automatically improve downstream RL training utility? Our experiments show that the answer is no. The physically calibrated twin is valuable, but its best role is not to replace the control-oriented surrogate; it is to regularize it. We then ask a second, pre-registered transferability question: which parts of the recipe transfer beyond bestest_air? The answer is

component-level rather than binary: the surrogate calibration pipeline transfers strongly, while the frozen controller does not transfer in a deployment-ready sense.

Contributions

1. A comfort-oriented v3 surrogate for direct supply-temperature control and RL rollout generation.
2. A physically informed v3.5 grey-box surrogate with explicit C_{zon} identified through Stage A/B/C inverse calibration.
3. A hybrid backend where v3 supplies the smooth RL dynamics and calibrated v3.5 acts as a frozen physical disagreement regularizer.
4. Controller-family-specific evidence: $\lambda_{temp}=0.10$ for thermostatic PPO, $\lambda_{temp}=0.00$ for HDRL and 17D MORL.
5. A pre-registered transferability analysis across three hydronic BOPTEST testcases, showing uniform surrogate-side transfer under full Stage A/B/C recalibration and regime-dependent frozen-controller transfer.
6. A reproducible Hou-and-Evins-style numerical audit with S1-S11 supplementary tables and real-data figures.

2. Related Work

2.1 Deep reinforcement learning for HVAC control

[TODO] Position PPO, hierarchical RL, safe RL, and MORL HVAC literature.

2.2 Surrogate and digital twin models for building energy

[TODO] Contrast predictive surrogate validity with closed-loop RL training utility.

2.3 Physics-informed and physics-guided machine learning

[TODO] Distinguish hard PINN constraints from the soft frozen regularizer used here.

2.4 BOPTEST benchmarking

[TODO] Describe BOPTEST, `bestest_air`, KPI definitions, and why BOPTEST RTE is the live evaluation layer.

3. Methodology

3.1 Control-oriented surrogate v3

The v3 surrogate is a direct-TSup, two-headed neural surrogate trained from BOPTEST trajectories. Its primary value is control-oriented smoothness: it gives PPO a stable local

environment with temperature and power heads. Its limitation is interpretability: it is a black-box dynamics model without explicit thermal capacitance.

3.2 Physically informed surrogate v3.5 and inverse calibration

The v3.5 surrogate introduces a physical backbone with explicit zone thermal capacitance C_{zon} . Stage A cleans and aligns telemetry, Stage B solves the inverse task by identifying C_{zon} on excitation windows, and Stage C calibrates residual heads while freezing the identified physical parameter.

3.3 Hybrid backend

The hybrid backend evolves the policy under v3 dynamics and computes an auxiliary disagreement penalty against frozen calibrated v3.5. Thus v3.5 does not enter the policy forward pass; it shapes the loss only.

$$L_{\text{total}} = L_{\text{PPO}} + \lambda_{\text{temp}} \|T_{\text{v3}} - T_{\text{v3.5}}\|^2 + \lambda_{\text{power}} \|P_{\text{v3}} - P_{\text{v3.5}}\|^2$$

Hybrid backend: v3 dynamics with frozen v3.5 physical regularizer

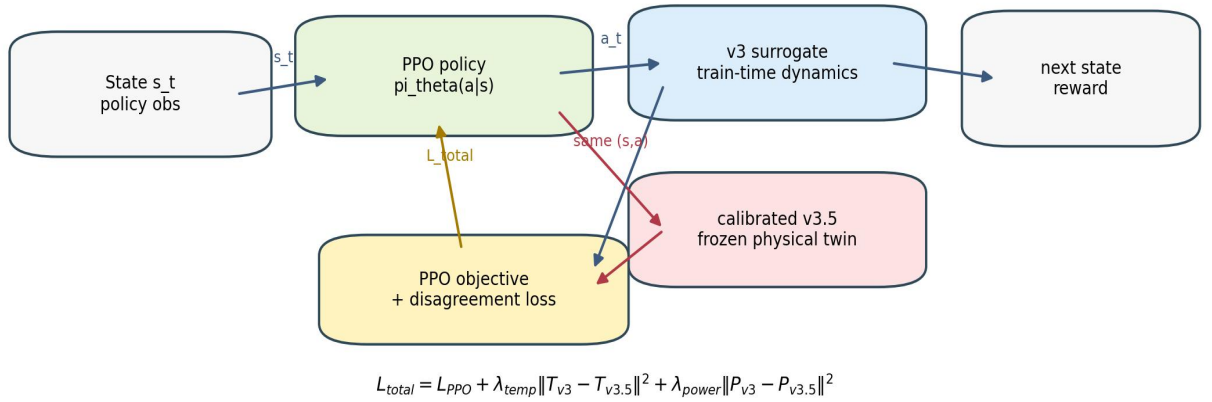


Figure 1. Hybrid backend schematic. The PPO policy rolls out through the smooth v3 surrogate while the calibrated v3.5 physical twin remains frozen and contributes only the disagreement regularizer.

3.4 Controller families

The controller stack contains the BOPTEST built-in PI reference, thermostatic PPO, HDRL, and 17D preference-conditioned MORL. Each family is evaluated under the same 15-min control protocol where possible.

4. Experimental Setup

4.1 Testbed

Blocks 1 and 2 use the BOPTTEST bestest_air testcase through boptest_rte HTTP. This choice is deliberate: the same serving layer used in experiments is also used in the speed benchmark. Block 3 extends the evaluation to three related single-zone hydronic testcases: bestest_hydronic_heat_pump, bestest_hydronic, and singlezone_commercial_hydronic. These transfer cases are adapter-mediated because their actuator interfaces are not literal direct supply-temperature interfaces.

4.2 Sample generation and preprocessing

The data generation, sample-size justification, split representativeness, Stage A preprocessing, scaling, input independence, and hyperparameters are reported in Supplementary Tables S1-S8.

4.3 Runtime characteristics

Backend	Steps/s	Median step ms	P95 step ms	Speed-up
boptest_rte_http	21.0	14.915	19.251	1.0x
v3_surrogate	4,626.3	0.165	0.273	220.2x
v35_calibrated_surrogate	2,399.5	0.359	0.495	114.2x
hybrid_v3_v35_surrogate	1,786.8	0.507	0.648	85.0x

Table 1. CPU throughput benchmark under the same 15-min control protocol. Source: reports/speed_benchmark_table.csv.

4.4 Evaluation protocol

All live-control results use 900 s control steps and 14-day monthly scenarios unless otherwise stated. Comfort is evaluated against the 21-24 C zone-temperature band. The main scalar safety metric is m_s, reported together with violation percentage and energy so that single-axis threshold decisions do not hide multi-objective trade-offs.

4.5 Reproducibility and audit trail

The project repository is https://github.com/Almaz-2001/HVAC_DRL_MORL.git. The reproducibility roadmap is maintained in roadmap.md, and the Block 3 pre-registration manifest is configs/block3_testcase_manifest.yaml. Three audit anchors are cited in the paper: MORL canonical pre-registration commit 93df9b364657ac77bbe3642e4bc277d1eb8a8b60; MORL post-N=5 falsification commit 62dc859d02f5f4a75fa4b55d8477c1d4e6206449; and Block 3 open/close commits 1861e48dc0eacb2e2c466ba0e0d03502d9185723 / b915bfc635c287dd1da907ce84ce44c81378edd5. A follow-up audit commit 7ada793bde6d9ae1483c389b813b11cc60bdec8a records the Block 3 close SHA in the manifest.

Experiments use BOPTEST RTE version 1.0.0-dev through the HTTP-Docker service. The local analysis environment used for this document reports Python 3.11.9 and PyTorch 2.10.0+cpu; training and BOPTEST control runs are configured for CPU execution only (configs/*/agent.yaml device=cpu, surrogate_device=cpu), with no GPU used. Seed handling explicitly propagates the seed through Python random, NumPy, PyTorch, PYTHONHASHSEED, and action/observation spaces where available. A fixed-checkpoint BOPTEST replay test was bit-identical across all 12 monthly scenarios, so canonical seed variance is attributed to RL training stochasticity rather than simulator nondeterminism.

5. Results I: Digital Twin Fidelity

5.1 Architecture comparison

Variant	Role	Physics	C_zon	Peak m_s	Typical m_s	Peak energy	Typical energy	Article position
v3	control-oriented direct-TSup surrogate	no	no	0.073	0.095	322.2	368.3	best pure control baseline
v35_calibrated	physics-oriented calibrated twin	yes	yes	1.046	1.102	556.2	580.5	best physical baseline, poor control surrogate
hybrid_I010	v3 dynamics regularized by calibrated v3.5	partial	regularizer-only	0.087	0.041	305.3	352.8	best verified compromise

Table 2. Architecture comparison of v3, calibrated v3.5, and hybrid_I010. Source: reports/hou_evins_architecture_justification_table.csv.

5.2 Predictive validity

Model	Horizon	RMSE_T	MAE_T	R2_T	RMSE_P	MAE_P
v3	1h	1.471	1.200	-0.982	1,227.6	815.5
v3	4h	1.579	1.274	-1.303	1,218.2	814.4
v3	8h	1.579	1.274	-1.355	1,201.7	807.1
v3	24h	1.558	1.258	-1.411	1,187.6	805.0
v3.5_calibrated	1h	0.321	0.220	0.905	688.8	480.7
v3.5_calibrated	4h	0.575	0.424	0.694	679.8	475.3
v3.5_calibrated	8h	0.638	0.476	0.616	664.7	468.0

v3.5_calibrated	24h	0.644	0.482	0.588	648.2	462.5
hybrid_I010	1h	1.471	1.200	-0.982	1,227.6	815.5
hybrid_I010	4h	1.579	1.274	-1.303	1,218.2	814.4
hybrid_I010	8h	1.579	1.274	-1.355	1,201.7	807.1
hybrid_I010	24h	1.558	1.258	-1.411	1,187.6	805.0

Table 3. Predictive validity over 1h/4h/8h/24h horizons. Source: reports/hou_evins_predictive_validity_table.csv.

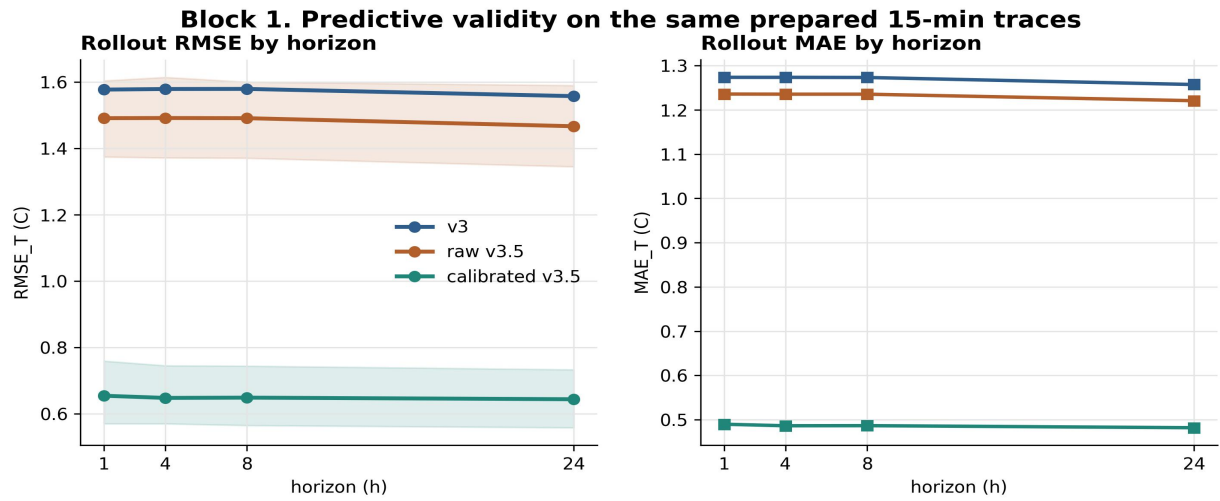


Figure 2. Block 1 predictive validity across 1h/4h/8h/24h horizons. The calibrated v3.5 twin improves long-horizon temperature prediction relative to raw v3.5 and establishes the physical twin's predictive value.

5.3 Fidelity-to-control gap

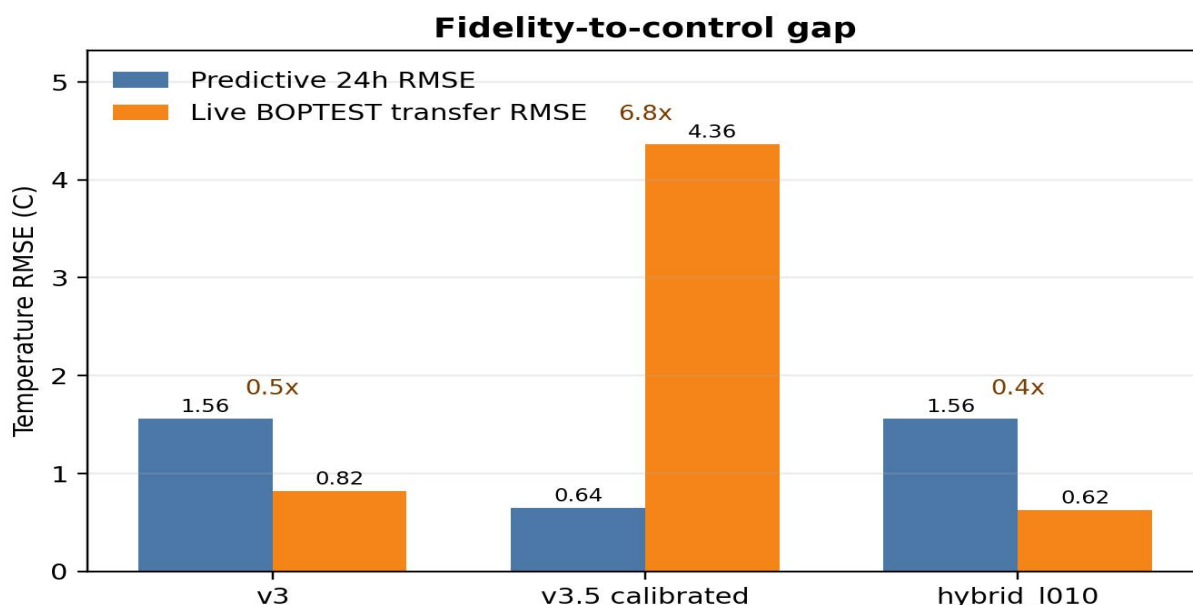


Figure 3. Fidelity-to-control gap. Predictive 24h RMSE alone does not determine live closed-loop utility: calibrated v3.5 is most predictive but has the largest live BOPTTEST transfer error when used as the standalone RL backend.

The calibrated v3.5 twin is predictive but not a good standalone RL training environment. Its live closed-loop transfer RMSE exceeds 4 C in the failure experiments, motivating the hybrid regularizer rather than direct replacement of v3. Residual distributions and programmatic 24h rollout traces are kept as supplementary/defense figures rather than main-body figures.

6. Results II: Control Performance

6.1 PI baseline

The BOPTTEST built-in PI controller is used as the reproducible standard reference, not as a custom-tuned strong PI baseline. Under yearly evaluation it has $m_s=0.910$, violation=63.59%, energy=104.07 kWh, and RMSE=3.395 C.

6.2 Negative control: direct v3.5 warm-start

The direct v3.5 warm-start is a negative control: better physical predictive fidelity does not by itself create a better RL backend. The evaluation summaries show worse closed-loop transfer than scratch in the relevant BOPTTEST KPIs, so v3.5 is used as a frozen regularizer rather than the rollout environment.

6.3 Thermostatic PPO with hybrid regularization

Model	Scenario	m_s	Violation %	Energy kWh	RMSE center C
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pure_v3	peak_heat_window	0.073	1.49	322.2	
pure_v3	typical_heat_window	0.095	4.39	368.3	
hybrid_I010	peak_heat_window	0.087	4.69	305.3	0.795
hybrid_I010	typical_heat_window	0.041	2.38	352.8	0.633

Table 4. Thermostatic pure-v3 versus hybrid_I010 BOPTEST KPIs.

Thermostatic PPO is the controller family that benefits most clearly from temperature disagreement regularization. The hybrid_I010 backend improves typical-window safety and energy relative to the pure-v3 controller while keeping the training loop in the fast in-process surrogate regime.

6.4 HDRL sensitivity to physical regularization

Variant	Scenario	m_s	Violation %	Energy kWh	RMSE center C
I000	peak_heat_window	0.180	6.10	329.6	0.751
I003	peak_heat_window	0.307	11.90	311.5	0.993
I005	peak_heat_window	0.418	20.98	298.1	1.303
I010	peak_heat_window	0.440	22.99	300.4	1.343
I000	typical_heat_window	0.234	3.12	385.1	0.691
I003	typical_heat_window	0.296	9.38	369.6	0.959
I005	typical_heat_window	0.512	27.38	354.5	1.491
I010	typical_heat_window	0.511	30.65	357.1	1.455

Table 5. HDRL lambda_temp sweep. Source: reports/block2_hdrl_lambda_sweep_summary.csv.

HDRL provides the main negative result for the temperature channel. As lambda_temp increases, the safety metric and violation rate degrade; the best HDRL configuration is lambda_temp=0. This prevents a generic claim that physical disagreement penalties are always beneficial.

6.5 MORL and Pareto front

Kind	Label	w comfort	w energy	m_s	Violation %	Energy kWh	RMSE C
morl_pareto	comfort_000_energy_100	0.00	1.00	1.588	86.76	0.3	8.359

mori_pareto	comfort_025_energy_075	0.25	0.75	0.173	11.39	220.0	0.912
mori_pareto	comfort_050_energy_050	0.50	0.50	0.193	12.92	237.0	0.939
mori_pareto	comfort_050_energy_050	0.50	0.50	0.103	6.77	239.3	0.769
mori_pareto	comfort_050_energy_050	0.50	0.50	0.141	9.34	238.2	0.879
mori_pareto	comfort_050_energy_050	0.50	0.50	0.187	11.97	236.7	0.895
mori_pareto	comfort_050_energy_050	0.50	0.50	0.310	24.05	212.8	0.985
mori_pareto	comfort_075_energy_025	0.75	0.25	0.057	3.08	249.5	0.700
mori_pareto	comfort_075_energy_025	0.75	0.25	0.087	5.47	247.0	0.716
mori_pareto	comfort_075_energy_025	0.75	0.25	0.117	7.82	234.9	0.769
mori_pareto	comfort_075_energy_025	0.75	0.25	0.160	9.42	253.1	0.903
mori_pareto	comfort_075_energy_025	0.75	0.25	0.276	20.37	216.9	0.909
mori_pareto	comfort_100_energy_000	1.00	0.00	0.032	1.49	260.6	0.631
mori_reference	legacy_canonical_080_020	0.80	0.20	0.099	4.87	248.6	0.725
baseline	pi_yearly_builtin			0.910	63.59	104.1	3.395

Table 6. MORL Pareto front and PI baseline. Source: reports/mori_pareto_front_table.csv.

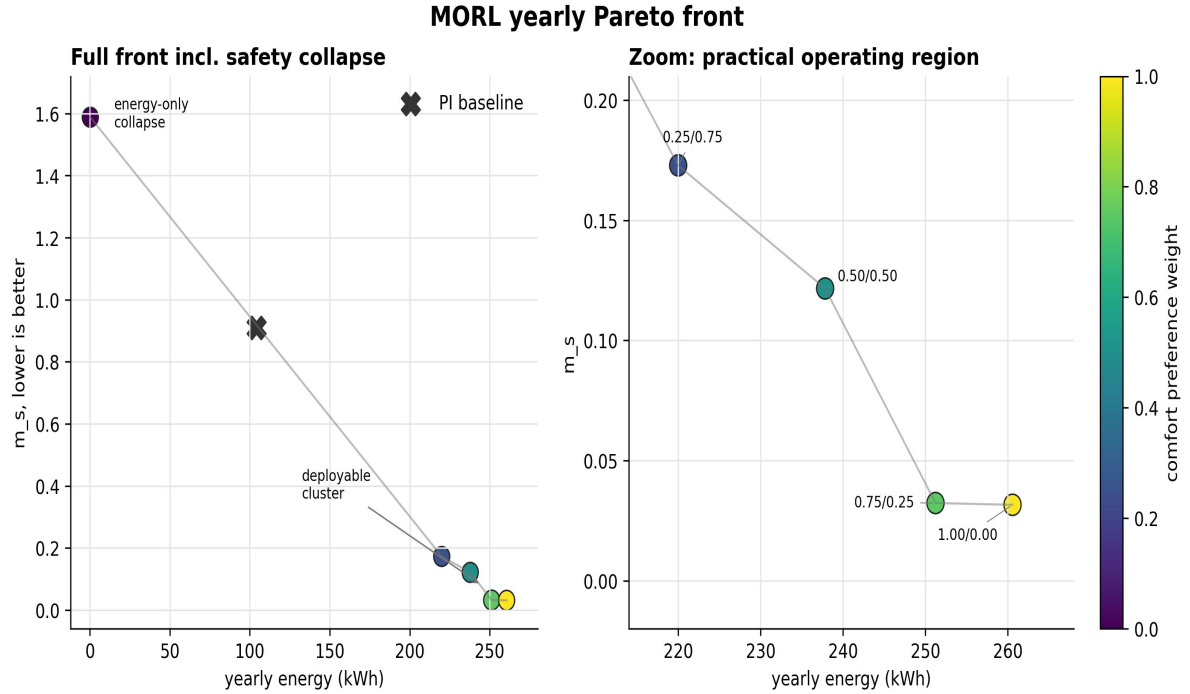


Figure 4. Block 2 MORL Pareto front with PI reference and canonical points. The energy-only endpoint demonstrates expected safety collapse; the canonical N=5 points provide variance-aware references rather than deployment-stability claims.

MORL N=5 falsification result

The neutral canonical $w=(0.50,0.50,0.00)$ closes at $m_s=0.187 \pm 0.078$ over five seeds ($\sigma/\mu=0.418$). The practical canonical $w=(0.75,0.25,0.00)$ improves the mean operating point to $m_s=0.139$ but remains high-variance at N=5. The replay test showed bit-identical BOPTEST yearly evaluation for a fixed checkpoint, so the variance is attributed to RL training stochasticity rather than simulator nondeterminism. The post-N=5 test falsified the action-saturation/seasonal-inversion hypothesis; the correct article claim is therefore that MORL is promising but not deployment-stable without future policy stabilization such as validation-based checkpoint selection or ensemble policy selection.

7. Results III: Transferability and Generalization

Block 3 reports a pre-registered transferability characterization of the hybrid recipe across three related BOPTEST hydronic testcases. Unlike Sections 5 and 6, which operate inside `bestest_air`, this section asks which components of the recipe transfer to a different actuator and envelope regime. The manifest was committed before Block 3 BOPTEST runs; result cells are appended rather than rewriting the pre-registration block.

7.1 Pre-registered protocol

The protocol crosses three testcases with recalibration regimes none, partial, and full. The target testcases are `bestest_hydronic_heat_pump`, `bestest_hydronic`, and

singlezone_commercial_hydronic. The primary live-control verdict is $m_s_RL \leq 1.25 \times m_s_PI$ under the same 15-min yearly evaluation protocol. Controller fine-tuning on the target-recalibrated surrogate is explicitly excluded from Block 3; it is reserved as future Block 4 work.

7.2 Actuator-interface adaptation

bestest_air exposes a direct supply-temperature style interface, but the hydronic testcases expose heat-pump, pump, valve, boiler, radiator, or commercial AHU variables. Therefore Block 3 transfer is adapter-mediated, not literal direct-TSup transfer. Each adapter mechanically maps the frozen policy's temperature-like action to documented hydronic setpoint and enable variables. The adapter is not learned and is smoke-tested before yearly evaluation.

7.3 Transfer matrix

Testcase	None	Full	m_s RL	m_s PI	Energy delta	Full RMSE gain	C_zon ratio	Interpretation
bestest_hydronic_heat_pump	FAIL	FAIL	0.665	0.464	-7.3%	60.2%	1.89x	Residential hydronic heat-pump transfer fails on controller safety but succeeds on full surrogate recalibration.
bestest_hydronic	FAIL	FAIL	0.976	0.750	-5.8%	87.4%	1.95x	Residential hydronic transfer replicates controller-side FAIL and surrogate-side PASS.
singlezone_commercial_hydronic	PASS	PASS	0.431	0.628	35.3%	87.8%	1.91x	Commercial stretch falsifies expected controller FAIL: safety passes but energy increases substantially; surrogate-side PASS and C_zon ratio remains near 1.9x.

Table 7. Block 3 transfer matrix. Controller verdicts use the pre-registered threshold $m_s_RL \leq 1.25 \times m_s_PI$. Full surrogate gains are diagnostic because the controller remains frozen by protocol. Source: [reports/block3_transfer_matrix.csv](#).

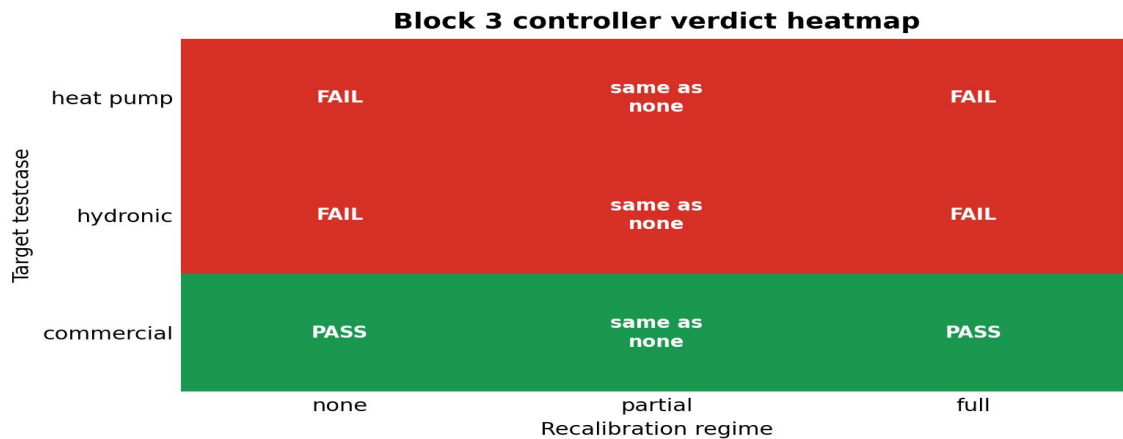


Figure 5. Block 3 transferability matrix. Red/green cells show the pre-registered controller verdict across three target testcases and three recalibration regimes. The partial regime is structurally identical to none for live-control KPIs because the controller remains frozen.

7.4 Testcase-level interpretation

The primary testcase, `bestest_hydronic_heat_pump`, fails controller transfer at `mode=none`: $m_s_RL=0.665$ versus $m_s_PI=0.464$ and threshold 0.579. Energy is 7.3% below PI, so the failure mode is comfort rather than energy. Full Stage A/B/C recalibration is strongly positive on the surrogate side, reducing $RMSE_T$ from 1.421 C to 0.565 C and re-identifying C_zon at 1.89x the `bestest_air` value.

The secondary testcase, `bestest_hydronic`, replicates this residential hydronic pattern. The frozen controller has $m_s_RL=0.976$ against threshold 0.938 and saves 5.8% energy versus PI, again failing on comfort. Full recalibration reduces $RMSE_T$ from 2.666 C to 0.335 C and re-identifies C_zon at 1.95x `bestest_air`.

The stretch testcase, `singlezone_commercial_hydronic`, breaks the binary failure pattern in a useful way. It is a threshold PASS on safety ($m_s_RL=0.431$ versus $m_s_PI=0.628$ and threshold 0.785), but it consumes 35.3% more energy than PI. Full recalibration again succeeds on the surrogate side, reducing $RMSE_T$ from 1.952 C to 0.238 C and identifying C_zon at 1.91x `bestest_air`.

7.5 Aggregate finding

Two patterns emerge. First, the surrogate component is uniformly transferable: full Stage A/B/C recalibration improves $RMSE_T$ by 60.2%, 87.4%, and 87.8% across the three hydronic testcases, while C_zon ratios are tightly clustered at 1.89x, 1.95x, and 1.91x. This supports testcase-portability of the Hou-and-Evins inverse calibration pipeline on the hydronic family.

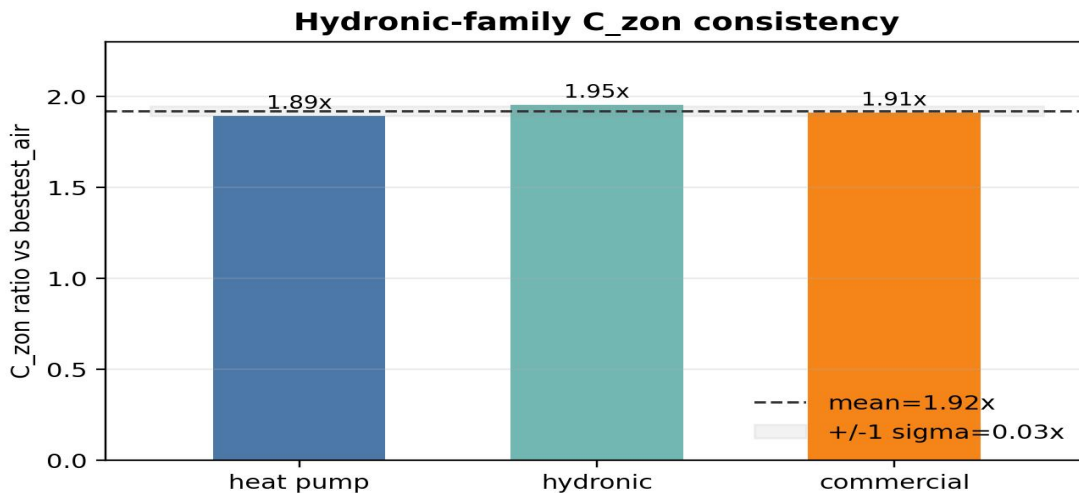


Figure 6. Block 3 C_zon consistency. Full Stage A/B/C recalibration re-identifies the hydronic-family thermal capacitance at approximately 1.9x the `bestestair` canonical value across all three target testcases.

Second, the frozen controller component is regime-dependent and not deployment-ready. On the two residential hydronic testcases it fails the safety threshold while saving energy. On the commercial hydronic testcase it passes the safety threshold but inflates energy by 35.3%. Both

modes reflect the same boundary: the transferred policy was trained on the direct-supply-temperature geometry of `bestest_air`, and a mechanical adapter cannot make it understand the target actuator's response curve.

7.6 Hypothesis closure and threshold caveat

H1_strong is falsified: frozen mode=none transfer is not deployment-ready across N=3 hydronic testcases. H2_medium is falsified by structural definition because partial recalibration updates only the surrogate while the live controller remains frozen. H3_weak is split: it is supported on the surrogate side and falsified on the controller side. The commercial cell also exposes a limitation of single-axis thresholds: it is a pre-registered threshold PASS on `m_s`, but not a deployment-ready pass because energy deteriorates by 35.3% versus PI.

8. Discussion

8.1 Predictive validity versus RL training utility

The central methodological result is that predictive validity and RL training utility are related but not equivalent. Predictive validation starts from held-out real BOPTEST states and measures whether a model can reproduce the next trajectory under known actions. Closed-loop RL training repeatedly visits states generated by the surrogate itself; small biases can therefore reshape the policy's experienced state distribution and produce qualitatively different control behavior. This explains why calibrated v3.5 can be a strong predictive twin and still fail as a standalone RL backend.

The hybrid backend resolves this mismatch by separating roles. The smoother v3 model remains the rollout environment, preserving a learnable control landscape. The calibrated v3.5 model acts only as a frozen soft regularizer, injecting physical information without forcing the policy to optimize inside the grey-box model's own closed-loop dynamics. This is the main reason the paper frames v3.5 as a regularizer rather than as a replacement simulator.

8.2 Controller-family specificity

The regularizer is not universally beneficial. Thermostatic PPO benefits from the physical anchor because its observation/action interface is low-dimensional and the disagreement penalty stabilizes the local temperature-power trade-off. HDRL and 17D MORL are more sensitive to the temperature disagreement channel and perform best with `lambda_temp=0` or power-only regularization. This controller-family specificity is important: the article does not claim a universal physics-guided penalty, but a measured role for physical disagreement that depends on the controller architecture.

The MORL findings sharpen this point. The 17D observation interface makes the Pareto sweep possible, but the canonical seed analysis remains high-variance at N=5. The failed action-saturation hypothesis is reported as a falsification result rather than hidden as noise. For deployment-oriented MORL, the next methodological layer should be policy stabilization, for

example validation-based checkpoint selection, early stopping, or seed ensembles. Those techniques are deliberately not applied post-hoc in this paper because they would change the pre-registered canonical evaluation protocol.

8.3 Transferability boundary

Block 3 decomposes transferability into a surrogate component and a controller component. The surrogate component transfers strongly: full Stage A/B/C recalibration improves RMSE_T by 60.2-87.8% across all three hydronic testcases, and the identified C_zon ratios are tightly clustered near 1.9x bestest_air. This consistency suggests that the inverse-calibration pipeline is portable across the tested hydronic family and that the hydronic cases are physically distinct from bestest_air rather than simple actuator aliases.

The controller component does not transfer in a deployment-ready sense. On the two residential hydronic cases, the frozen controller saves energy but fails the comfort/safety threshold. On the commercial hydronic case, it passes the pre-registered m_s threshold but uses 35.3% more energy than PI. These two failure modes have the same root: the transferred policy was trained for direct supply-temperature geometry and a mechanical adapter cannot teach it the target actuator response curve. The natural Block 4 experiment is therefore controller fine-tuning on the target-recalibrated surrogate, not further surrogate calibration alone.

8.4 Threats to validity

Several limits remain. The bestest_air controller evidence uses one weather file and targeted sensitivity analysis rather than full hyperparameter optimization. HDRL is single-seed. MORL canonical points use N=5 but remain high-variance, so MORL is reported as promising rather than deployment-stable. Block 3 uses three related hydronic testcases, not arbitrary building archetypes, multi-zone systems, or climate zones. The 85x speed-up is measured against the practical BOPTEST RTE HTTP-Docker deployment rather than bare FMU evaluation; a direct-FMU benchmark is logged as a platform-limited reproducibility item.

The Block 3 threshold also has a known limitation. The pre-registered verdict uses $m_{s_RL} \leq 1.25 \times m_{s_PI}$, which is useful for auditability but can mask individual KPI deterioration. The commercial hydronic testcase is the example: it is a threshold PASS on m_s but not a deployment-ready pass because energy increases by 35.3% versus PI. Future transferability protocols should use a tiered verdict: composite safety threshold plus per-KPI floors for energy and violation rate.

9. Conclusion

This paper tests a common assumption in physics-informed RL for buildings: that a more predictive physical twin should be the better RL environment. On BOPTEST bestest_air, the answer is negative. Calibrated v3.5 improves long-horizon predictive fidelity, but direct use as the RL backend fails in live closed-loop control. The successful role of the calibrated twin is

narrower and more useful: it serves as a frozen soft regularizer for a smoother v3 rollout backend.

The resulting hybrid recipe gives a reproducible control improvement for thermostatic PPO and preserves the speed advantage required for RL training. It also clarifies limits. HDRL rejects the temperature disagreement channel, and MORL produces a usable 17D Pareto front but remains high-variance at $N=5$ canonical seeds. These are not hidden weaknesses; they define where the method works and where additional stabilization is required.

The transferability block extends the contribution beyond `bestest_air`. Across three hydronic BOPTTEST testcases, the Stage A/B/C inverse surrogate-calibration pipeline transfers robustly, with 60.2-87.8% RMSE_T improvement and consistent C_{zon} re-identification near 1.9x `bestest_air`. The frozen controller component does not transfer in a deployment-ready sense: residential hydronic cases fail comfort, while the commercial case passes the scalar threshold only by accepting a large energy penalty. The final conclusion is therefore component-level: the surrogate physics representation transfers; the controller-adapter interface is the bottleneck.

The immediate next experiment is not another surrogate diagnostic, but target-specific controller fine-tuning on the target-recalibrated surrogate under a tiered comfort-energy transfer criterion. That experiment is intentionally left outside the current pre-registered scope so that the present paper can report the falsifications and boundaries without moving the goalposts.

Data availability

All numerical values are sourced from CSV artifacts under `reports/` and `outputs/`. The figure-source manifest is `reports/article_real_figures_manifest.csv`.

Supplementary Material: Hou and Evins Numerical Artifacts

The following tables summarize the eleven article-facing numerical artifacts. The complete machine-readable versions remain in reports/hou_evins_*.csv.

S1. Sample generation (reports/hou_evins_sample_generation_table.csv)

dataset_id	rows	step_sec	controller_or_policy_mix	intended_role	article_status
v3_hourly_direct_tsup	51,200.0	3,600.0	cool,heat,mixed,random	broad seasonal direct-TSup control surrogate training corpus	control-oriented base corpus
v35_prepared_15min_bootstrap	10,744.0	900.0	hdrl,thermostatic	15-minute closed-loop bootstrap dataset for initial rollout-oriented calibration	controller-biased bootstrap corpus
v35_collected_15min_exploration	48,384.0	900.0	cool,heat,mixed,pulse,random,thermostatic_noise	broader 15-minute exploration dataset for transfer-stability analysis	exploratory robustness corpus

S2. Sample-size justification

(reports/hou_evins_sample_size_justification_table.csv)

dataset_id	rows	primary_metric_name	primary_metric_value	decision	justification
v3_hourly_direct_tsup	51,200.0	thermostatic_m_s_mean	0.084	retain	Broadest seasonal-policy coverage and strongest pure-control baseline; retained as the control-oriented surrogate corpus.
v35_prepared_15min_bootstrap	10,744.0	calibrated_rollout_24h_rmse_c	0.644	retain	Smallest 15-minute corpus that already supports explicit C _{zon} identification and the canonical calibrated v3.5 backend without additional live collection cost.
collected_15min_exploration	48,384.0	h4_rmse_c	1.115	reject_as_canonical	Much larger 15-minute corpus required new BOPTEST collection time, but still underperformed the prepared bootstrap corpus on rollout-safety metrics.

collected_15min_focus	24,192.0	h4_rmse_c	0.921	reject_as_canonical	More targeted heating-focused recollection reduced cost relative to the full collected corpus, but still did not beat the prepared bootstrap corpus on 1-hour safety fidelity.
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S3. Split representativeness

(reports/hou_evins_split_representativeness_table.csv)

pipeline	split_mode	train_rows	val_rows	representativeness_assessment	assessment_note
v3_hourly_control_bacbone	row_index_contiguous_80_20	40,960.0	10,240.0	limited	Validation keeps all control policies but only covers the tail of the ordered corpus (autumn-only by season), so final claims rely on external BOPTEST benchmarks, not this split alone.
v35_inverse_canonical_power_head_only	episode_id_80_20_plus_excitation_subselection	8,058.0	2,686.0	moderate	Both heating scenarios are present in train and val; val is smaller and thermostatic-only, but the final branch is still tested downstream on live BOPTEST transfer.
hybrid_thermostatic_downstream	no_internal_dataset_split			external_tested	Downstream control claims are based on explicit out-of-training BOPTEST evaluation rather than a held-out supervised split, which is the more relevant test for controller utility.

S4. Stage A processing (reports/hou_evins_stage_a_processing_table.csv)

stage_a_operation	implementation	parameter_or_rule	purpose
latency compensation search	_undo_delay_array over candidate lags	lag in [0, max_latency_search]	undo thermostat/telemetry delay before calibration
temperature bias removal	median residual offset subtraction	bias_est = median(t_obs_next - t_pred)	avoid fitting static sensor bias as building physics
power affine normalization	least-squares fit $p_{\text{obs}} \approx \text{scale} \cdot$	scale_est, bias_p_est from linear	separate systematic meter scaling

	p_pred + bias	solve	from dynamic error
rolling denoise	rolling median + exponential moving average	smooth_window = 5	stabilize downstream inverse calibration
causal delta recomputation	delta_t = t_zone_next - t_zone after preprocessing	derived from corrected current and next temperature	avoid propagating artifact-driven temperature-history features

S5. Feature significance and encoding (reports/hou_evins_feature_justification_table.csv)

variant	obs_ablation	power_feature_mode	t_zone_feature_mode	peak_m_s	typical_m_s	decision
with_power_raw	none	raw	raw	1.270	1.290	reference-only
no_power	no_power	raw	raw	1.281	1.290	rejected
no_delta_t	no_delta_t	raw	raw	1.036	1.121	retained-as-direction
no_power_no_delta_t	no_power_no_delta_t	raw	raw	1.062	1.123	rejected
no_prev_action	no_prev_action	raw	raw	1.259	1.280	rejected
no_temp_history	no_temp_history	raw	raw	1.085	1.149	rejected
causal_smooth	none	raw	raw	1.129	1.187	rejected
no_delta_t_powerlog	no_delta_t	clipped_log	raw	1.090	1.120	retained-intermediate
no_delta_t_powerlog_t_zone	no_delta_t	clipped_log	comfort_centered	1.046	1.102	retained-diagnostic

S6. Scaling (reports/hou_evins_scaling_table.csv)

variable	context	scaling_method	parameters	justification
surrogate_t_zone	v3/v3.5 surrogate input	affine min-max to [-1, 1]	min=15 C; max=35 C	Keeps the learned thermal state within the training support used by the direct-TSup surrogate.
surrogate_t_amb	v3/v3.5 surrogate input	affine min-max to [-1, 1]	min=-10 C; max=40 C	Normalizes outdoor-air temperature for the heat and power heads.
hour	v3/v3.5 surrogate input and TSup controller observation	cyclic sine/cosine	$\sin(2\pi \cdot \text{hour}/24)$; $\cos(2\pi \cdot \text{hour}/24)$	Avoids discontinuity between hour 23 and hour 0.

day	v3/v3.5 surrogate input and TSup controller observation	cyclic sine/cosine	$\sin(2\pi \cdot \text{day}/365)$; $\cos(2\pi \cdot \text{day}/365)$	Represents seasonal phase without an artificial year-end jump.
a0_raw	direct supply-temperature action	native action domain	[-1, 1] maps to 18..35 C supply temperature	Preserves the PPO action coordinate while keeping physical supply-temperature bounds explicit.
a1_raw	direct fan action	native action domain	[-1, 1] maps to 0..1 fan command	Keeps fan modulation aligned with the controller action range.
p_total	TSup controller observation, raw mode	affine min-max to [-1, 1]	min=0 W; max=5500 W	Raw power normalization for ablation baselines.
p_total	canonical TSup controller observation	clipped log transform to [-1, 1]	clip at 2500 W; $2 \cdot \log_{10}(p) / \log_{10}(2500) - 1$	Compresses high-power tails and improved transfer in the canonical hybrid branch.
delta_t	TSup controller observation, raw mode	linear clipping	clip($\Delta T / 5$ C, -1, 1)	Baseline delta-temperature history encoding.
delta_t	MORL 17D TSup-style observation	causal smooth tanh	clip to +/-1.5 C; $\tanh(\Delta T / 1.25)$	Reduces action discontinuities from noisy short-step temperature changes.
controller_t_zone	TSup controller observation, raw mode	affine min-max to [-1, 1]	min=5 C; max=40 C	Wide physical range for controller-side temperature observation.
controller_t_zone	TSup controller observation, comfort-centered ablation	comfort-centered clipping	clip($(T_{\text{zone}} - 22.5 \text{ C}) / 3$ C, -1, 1)	Ablation tested a comfort-local coordinate, but canonical thermostatic hybrid uses raw mode.
power_head_output	v3 surrogate output	Softplus multiplied by P_MAX	P_MAX=5500 W	Guarantees non-negative predicted HVAC power.
C_zon	v3.5 physics backbone	positive reparameterization	$\text{softplus}(\log_{10}(C_{\text{zon}})) + c_{\text{min}}$; $c_{\text{min}} = 5e4 \text{ J/K}$	Constrains the identified thermal capacitance to physically meaningful positive values.

S7. Input independence (reports/hou_evins_input_independence_table.csv)

feature_i	feature_j	pearson_r	normalized_mutual_info	interpretation
t_zone	t_amb	-0.145	0.062	low_to_moderate_dependency

t_zone	hour_sin	-0.363	0.058	low_to_moderate_dependency
t_zone	hour_cos	-0.003	0.056	low_to_moderate_dependency
t_zone	day_sin	-0.144	0.050	low_to_moderate_dependency
t_zone	day_cos	0.112	0.051	low_to_moderate_dependency
t_zone	prev_action_a0	0.130	0.111	low_to_moderate_dependency
t_zone	prev_action_a1	0.148	0.047	low_to_moderate_dependency
t_zone	power_history_proxy	0.245	0.051	low_to_moderate_dependency
t_zone	delta_t_history	-0.179	0.042	low_to_moderate_dependency
t_amb	hour_sin	-0.308	0.084	low_to_moderate_dependency
t_amb	hour_cos	-0.366	0.084	low_to_moderate_dependency
t_amb	day_sin	0.025	0.229	low_to_moderate_dependency
t_amb	day_cos	0.013	0.195	low_to_moderate_dependency
t_amb	prev_action_a0	-0.801	0.248	moderate_dependency_expected_from_physics_or_time
t_amb	prev_action_a1	-0.058	0.122	low_to_moderate_dependency
t_amb	power_history_proxy	-0.638	0.227	moderate_dependency_expected_from_physics_or_time
t_amb	delta_t_history	0.061	0.016	low_to_moderate_dependency
hour_sin	hour_cos	-0.000	0.573	moderate_dependency_expe

				cted_from_physics_or_time
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S8. Training hyperparameters

(reports/hou_evins_training_hyperparams_table.csv)

param	value	source_file	justification
active_v3_training_entrypoint	surrogate/train_surrogate_backbone.py	surrogate/train_surrogate_backbone.py	Canonical comfort-oriented direct-TSup v3 backbone training script currently present in active contour.
legacy_train_surrogate_py	not present in active contour	surrogate/train_surrogate.py	The former generic entrypoint is not part of the current active contour; v3 training is reported through train_surrogate_backbone.py.
epochs	500	surrogate/train_surrogate_backbone.py	Default CLI value; upper budget constrained by early stopping.
batch_size	256	surrogate/train_surrogate_backbone.py	Default DataLoader batch size used for stable mini-batch estimates on the BOPTEST corpus.
learning_rate	1e-3	surrogate/train_surrogate_backbone.py	Default AdamW learning rate for the v3 backbone.
hidden_dim	64	surrogate/train_surrogate_backbone.py	Default hidden width used by RCNeuralODEv2 heat and power heads.
optimizer	AdamW	surrogate/train_surrogate_backbone.py	Weight-decayed Adam optimizer stabilizes the black-box residual heads without changing the physical inputs.
weight_decay	1e-4	surrogate/train_surrogate_backbone.py	Explicit optimizer regularization in the canonical v3 training loop.
scheduler	CosineAnnealingLR(T_max=epochs)	surrogate/train_surrogate_backbone.py	Smooth learning-rate decay over the configured epoch budget.
early_stopping	patience=30; improvement_threshold=1e-6	surrogate/train_surrogate_backbone.py	Stops training when the selected validation metric no longer improves.
checkpoint_metric	val_loss default; rollout_rmse optional	surrogate/train_surrogate_backbone.py	Default keeps one-step supervised fidelity; rollout_rmse was added for rollout-aware selection when explicitly requested.
gradient_clip_norm	1.0	surrogate/train_surrogate_backbone.py	Limits unstable updates from multi-

		py	step rollout loss.
multi_horizons	[2, 4]	surrogate/train_surrogate_backbone.py	Default short rollout horizons for one-step-plus-local-rollout training.
loss_lambda_temp	1.0	surrogate/train_surrogate_backbone.py	Primary temperature fidelity term.
loss_lambda_power	0.1	surrogate/train_surrogate_backbone.py	Power-head term is normalized by P_MAX and weighted below the temperature term.
loss_lambda_multi	0.5	surrogate/train_surrogate_backbone.py	Balances local rollout consistency against one-step fit.
loss_lambda_physics	0.05	surrogate/train_surrogate_backbone.py	Penalizes implausibly large one-step temperature jumps.

S9. Architecture justification (reports/hou_evins_architecture_justification_table.csv)

variant	role	explicit_physics	explicit_c_zon	peak_control_m_s	typical_control_m_s	article_position
v3	control-oriented direct-TSup surrogate	no	no	0.073	0.095	best pure control baseline
v35_calibrated	physics-oriented calibrated twin	yes	yes	1.046	1.102	best physical baseline, poor control surrogate
hybrid_l010	v3 dynamics regularized by calibrated v3.5	partial	regularizer-only	0.087	0.041	best verified compromise

S10. Targeted sensitivity (reports/hou_evins_targeted_sensitivity_table.csv)

sensitivity_axis	tested_values	selection_metric	winner	numerical_reason
observation_ablation	none,no_power,no_delta_t,no_power_no_delta_t,no_prev_action,no_temp_history	thermostatic transfer/control stability	no_delta_t	Best direct v3.5 diagnostic row was no_delta_t with peak m_s=1.0361 and typical m_s=1.1208.
feature_encoding	raw,causal_smooth,clipped_log,comfort_centered	reduced transfer-gap without reopening full model search	power=clipped_log,t_zone=raw,delta=removed	Encoding tests were evaluated through direct v3.5 transfer diagnostics and then reused in the canonical hybrid branch instead of running a global architecture search.

hybrid_lambda_temp_disagree	0.05,0.10,0.15	mean thermostatic m_s across peak/typical	l010	lambda 0.010 achieved the best mean m_s among the tested values (0.0639).
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S11. Predictive validity (reports/hou_evins_predictive_validity_table.csv)

model	horizon	RMSE_T	MAE_T	RMSE_P	MAE_P	source_note
v3	1h	1.471	1.200	1,227.6	815.5	direct-TSup v3 primary dynamics
v3	4h	1.579	1.274	1,218.2	814.4	direct-TSup v3 primary dynamics
v3	8h	1.579	1.274	1,201.7	807.1	direct-TSup v3 primary dynamics
v3	24h	1.558	1.258	1,187.6	805.0	direct-TSup v3 primary dynamics
v3.5_calibrated	1h	0.321	0.220	688.8	480.7	calibrated v3.5 physical backbone and power head
v3.5_calibrated	4h	0.575	0.424	679.8	475.3	calibrated v3.5 physical backbone and power head
v3.5_calibrated	8h	0.638	0.476	664.7	468.0	calibrated v3.5 physical backbone and power head
v3.5_calibrated	24h	0.644	0.482	648.2	462.5	calibrated v3.5 physical backbone and power head
hybrid_l010	1h	1.471	1.200	1,227.6	815.5	v3 primary rollout dynamics; calibrated v3.5 disagreement regularizer used during control training
hybrid_l010	4h	1.579	1.274	1,218.2	814.4	v3 primary rollout dynamics; calibrated v3.5 disagreement regularizer used during control training
hybrid_l010	8h	1.579	1.274	1,201.7	807.1	v3 primary rollout dynamics; calibrated v3.5 disagreement regularizer used during control training
hybrid_l010	24h	1.558	1.258	1,187.6	805.0	v3 primary rollout dynamics; calibrated v3.5 disagreement regularizer used during control training

						control training
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References

[TODO] Populate from bibliography.bib: Hou and Evins (2024), BOPTEST, PPO, PINN, DRL-HVAC, MORL-HVAC, and physics-guided ML references.